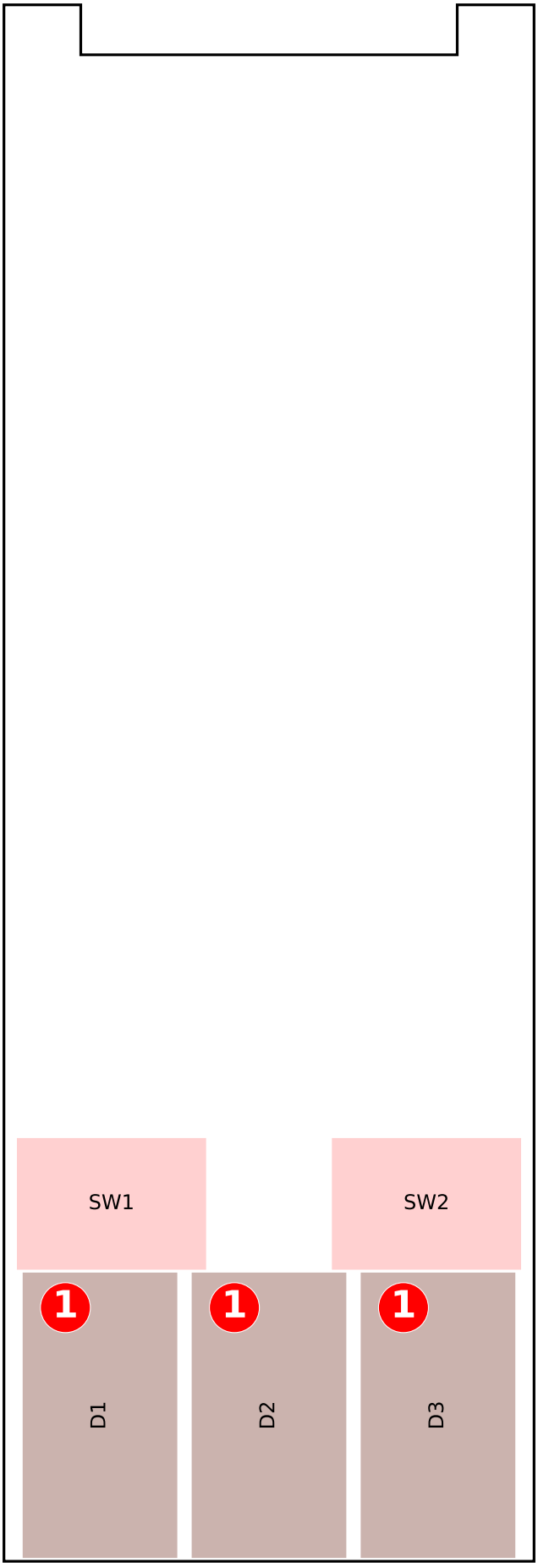
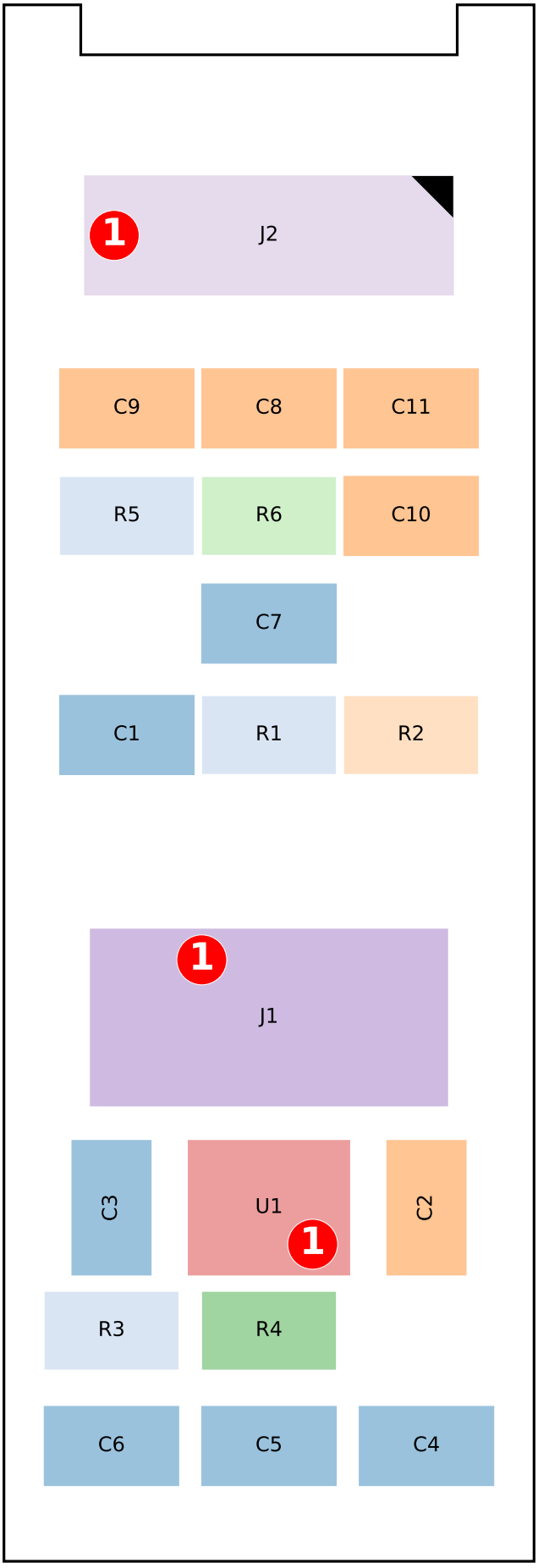
ui board: TOP side

black bar = diode cathode; red + = capacitor +; red 1 = pin 1; black corner = hand-fitted; same colour = same value (values page)



ui board: BOTTOM side (seen from below, mirrored)

black bar = diode cathode; red + = capacitor +; red 1 = pin 1; black corner = hand-fitted; same colour = same value (values page)



ui board: parts by value

100n 50V (0805)	x6	bottom	-	C1, C3, C4, C5, C6, C7 CC0805KRX7R9BB104 C49678
10k (0805)	x3	bottom	-	R1, R3, R5 0805W8F1002T5E C17414
1u 50V (0805)	x5	bottom	-	C2, C8, C9, C10, C11 CC0805KKX7R9BB105 C91185
20k (0805)	x1	bottom	-	R2 0805W8F2002T5E C4328
330R (0805)	x1	bottom	-	R4 0805W8F3300T5E C17630
390k (0805)	x1	bottom	-	R6 0805W8F3903T5E C17656
74AHCT1G125 (SOT-23-5)	x1	bottom	pin 1	U1 SN74AHCT1G125DBVR C7484
B3U-1000P (SW)	x2	top	-	SW1, SW2 B3U-1000P C231329
FPC-05F-8PH20 (FPC)	x1	bottom	pin 1	J1 FPC-05F-8PH20 C2856797
OLED (OLED)	x1	bottom	pin 1	J2 QG-6428TSWCG02
WS2812B-MINI-V3J (LED)	x3	top	pin 1	D1, D2, D3 WS2812B-MINI-V3J C22371519

ui board: key

Marks on the drawing

Black bar across a diode: the cathode end. Put the band printed on the diode on this side.

Red +: the + pad of an electrolytic or tantalum capacitor.

Red dot with 1: pin 1 of an IC, transistor, LED, connector or module.

No mark: resistors, ceramic capacitors, inductors, fuses, crystals, pin strips and push buttons can go either way round. Bidirectional TVS diodes (part numbers ending in CA) have no polarity either.

Black top-right corner: hand-fitted part (not placed by JLCPCB), e.g. a socketed module or a terminal JLC has no stock of.

The bottom page is mirrored: it shows the board as you see it after flipping it over.

Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pins 1 and 2 are on the opposite side.

SOT-23-5 / TSOT-23-6 / SC-70 / SO / QFN: a dot or a bevelled edge on the body marks pin 1.

Addressable RGB LEDs: one corner of the body or lens is marked (a cut corner or a dot). Which pin it marks differs between makers: check the part's datasheet before soldering, it is easy to misread.

0805 LEDs: pin 1 is the cathode, usually a green line or a T on the underside.

Diodes: the band is the cathode.

Electrolytic capacitors: the stripe on the can marks the - side; the + pad is the other one.

Modules and connectors: match the pin-1 dot to the module's pin-1 marking or to the drawing in its datasheet; for modules on sockets, check the outline (e.g. USB connector towards the board edge).

Tools

0805 parts and SOT-23: a 3-5x magnifier lamp or a 10x loupe to read markings and check joints.

SC-70, QFN and fine-pitch connectors: a USB microscope helps to see bridges and pin-1 marks.

Flux, fine solder (0.5 mm) and a chisel or fine conical tip; tweezers for placing.